GGG-R-425b
February 20, 1974
SUPERSEDING
Fed. Spec. GGG-R-425a
June 10, 1963

FEDERAL SPECIFICATION

RIVET SET, PNEUMATIC TOOL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers rivet sets for use in portable power hammers or riveters for driving and setting rivets.

1.1.1 Federal specification coverage. This specification does not cover all varieties of rivet sets that are commercially available but is intended to include only those varieties generally purchased by the Federal Government.

1.2 Classification.

1.2.1 Types, classes, styles, and shapes. Rivet sets shall be of the following types, classes, styles, and shapes as specified (see 6.2):

Type I - Rivet set for hammers.
  Class 1 - Flush head.
  Class 2 - Button head.

Type II - Rivet set for riveters.
  Class 1 - Flush head.
    Style A - Standard shank.
    Style B - Parker taper shank.
  Class 2 - Button head.
  Class 3 - Universal head.
    Style A - Short shank.
    Style B - Parker taper shank.
      Shape O - Offset body.
      Shape S - Straight body.
  Class 4 - Brazier head.
  Class 5 - Modified brazier head.
  Class 6 - Round head.
    Stype B - Parker taper shank.
      Shape O - Offset body.
      Shape S - Straight body.

1.2.2 Size. The size of the rivet set shall correspond to the "rivet size" as shown in the applicable tables and as specified (see 6.2). Size is not applicable to the type I, class 1 rivet set since its use is not limited to specific rivet size.

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-P-40 - Packaging and Packing of Hand Tools.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards, and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is specified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

American National Standards Institute (ANSI), Inc., Standard:

B46.1 - Surface Texture.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations herein are descriptive and not restrictive and are not intended to preclude the purchase of rivet sets otherwise conf ng to this specification.

3.2 Material. Rivet sets shall be made from high grade tool s el properly heat treated to conform to the requirements, tests, and examinations specified herein.

3.3 Shank, body, and cupping. The shank, body, and cupping of the rivet sets shall be of the size and shape as specified herein. The term "cupping" shall apply to the driving face of flush rivet sets and to the cups and cup rims of other rivet sets.

3.4 Finish. The shank of the rivet sets shall have a surface roughness equivalent to 125 microinches or better, the cupping shall have 8 microinches or better. The body may be unfinished but shall be coated with black oxide followed with a nondrying petroleum oil.

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3.5 Marking. The body or shoulder of each rivet set shall be legibly and permanently marked with numbers or symbols to indicate the size and type of rivet and the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.6 Dimensional tolerances. Unless otherwise specified, dimensional tolerances shall be ±1/64 on fractions, ±0.010 on decimals, and ±5° on angles. Unless otherwise specified, all dimensions are in inches.

3.7 Type I rivet set for hammers. The body cross section of the type I rivet set shall be octagonal in shape and all areas shall have a hardness of 46 to 50 on the Rockwell C scale.

3.7.1 Class 1, flush head rivet set. The type I, class 1 rivet set shall be similar to and conform to the dimensions shown on figure 1.

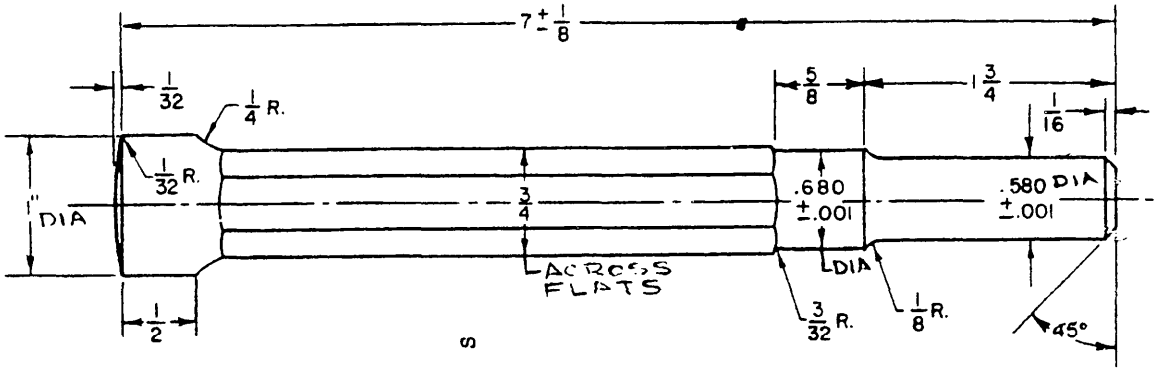


FIGURE 1. Type I, class 1 flush-head rivet set for hammers.

3.7.2 Class 2, button head rivet set. The type I, class 2 rivet set shall be similar to figure 2 and conform to the dimensions in figure 2 and on table I for the size rivet specified (see 6.2).

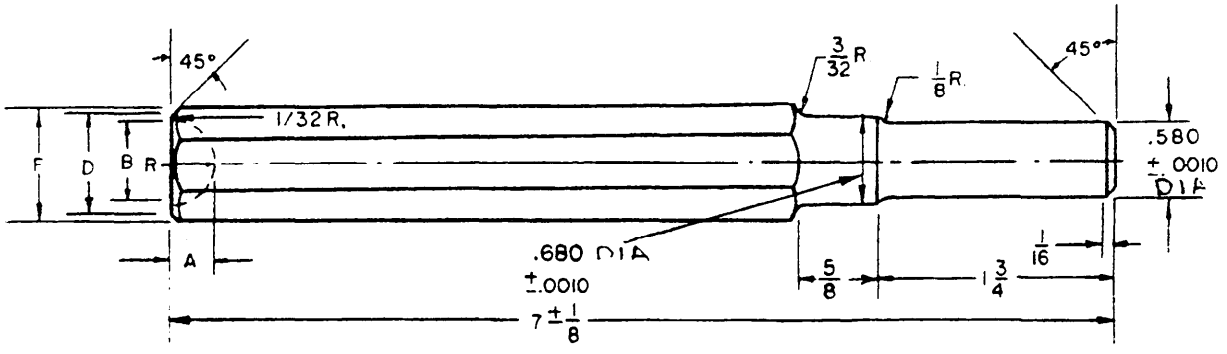


FIGURE 2. Type I, class 2 button-head rivet set for hammers.

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TABLE I. Dimensions for type I, class 2 button-head rivet sets

Rivet size	A Depth of cupping	B Diameter of cupping	D Diameter of cup rim	F Body size (across flats)	R Radius of cup
1/4	11/64	29/64	47/64	3/4	15/64
3/8	1/4	11/16	13/16	7/8	23/64

3.8 Type II rivet set for riveters. The body cross section of the type II rivet set shall be round and all areas shall have a hardness of 46 to 50 on the Rockwell C scale.

3.8.1 Class 1, flush head rivet set.

3.8.1.1 Style A, standard shank. The type II, class 1, style A rivet set shall be similar to figure 3 and conform to the dimensions on figure 3 and in table II for the size rivet specified (see 6.2).

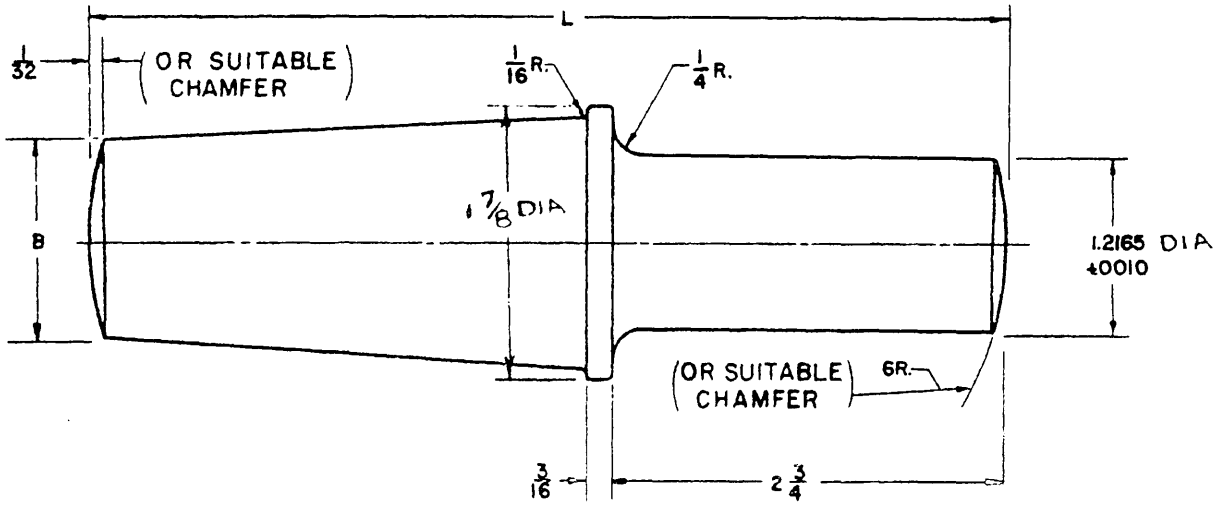


FIGURE 3. Type II, class 1, style A, standard-shank, flush-head rivet set.

TABLE II. Dimensions for type II, class 1, style A, standard shank, flush-head rivet sets

Rivet size	B Diameter of cupping	L Overall length ±1/8
1/2	3/4	4-1/2
3/4	1-11/32	4-1/2
1	1-3/4	4-1/2
1-1/4	2-1/16	4-3/4

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3.8.1.2 Style B, Parker taper shank. The type II, class 1, style B rivet set shall be similar to figure 4 and conform to the dimensions on figure 4 and in table III for the rivet size specified (see 6.2).

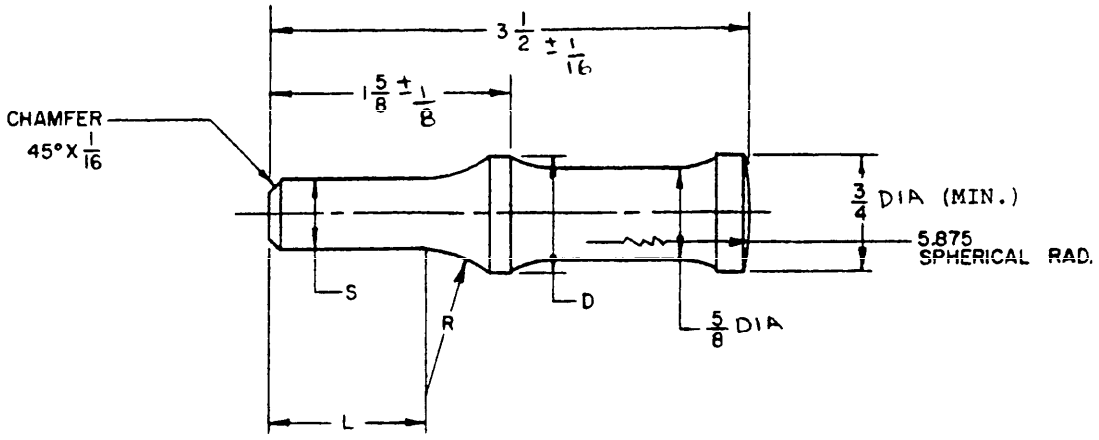


FIGURE 4. Type II, class 1, style B, Parker taper shank flush-head rivet set.

TABLE III. Dimensions for type II, class 1, style B, Parker taper shank, flush-head rivet sets

Rivet size	S Shank diameter +.0005 -.0015	L Shank length	R Shank radius	D Flange diameter
1/4	.401	1.16	.50	3/4
3/8	.498	1.10	.75	13/16

3.8.2 Class 2, button head. The type II, class 2 rivet set shall be similar to figure 5 and conform to the dimensions on figure 5 and in table IV for the rivet size specified (see 6.2).

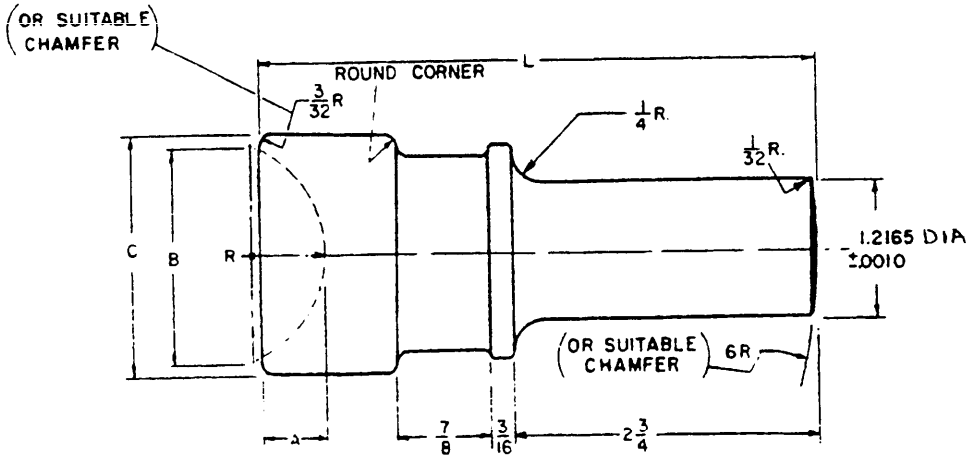


FIGURE 5. Type II, class 2 button-head rivet set.

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TABLE IV. Dimensions for type II, class 2, button-head rivet sets

Rivet size	A Depth of cupping	B Diameter of cupping	R Radius of cupping	L Overall length	C Head diameter
3/8	15/64	19/32	5/16	4-1/2	1-7/8
1/2	5/16	25/32	13/32	4-1/2	1-7/8
5/8	13/32	1	33/64	4-1/2	1-7/8
3/4	31/64	1-3/16	39/64	4-1/2	1-15/16
7/8	35/64	1-3/8	45/64	4-5/8	2
1	41/64	1-19/32	13/16	4-3/4	2-1/4
1-1/8	45/64	1-3/4	57/64	4-7/8	2-3/8
1-1/4	51/64	2	1- 1/32	5	2-5/8

3.8.3 Class 3, universal head. The type II, class 3 universal head rivet set shall have a cupping conforming to figure 6 and the dimensions shown in table V for the size rivet set specified (see 6.2).

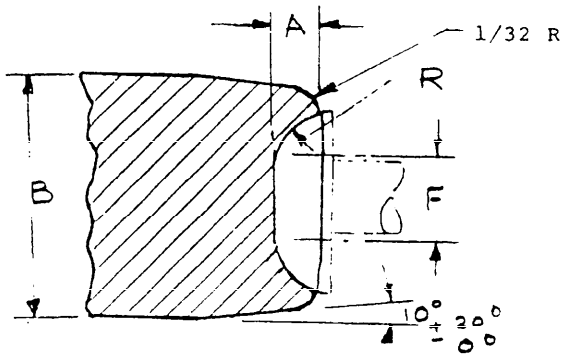


FIGURE 6. Cupping for type II, class 3 rivet set.

TABLE V. Cupping dimensions for type II, class 3 rivet sets

Rivet size	A Depth of cupping +.002 -.000	F Cupping flat +.004 -.000	R Radius of cupping +.001 -.000	B Body diameter
1/16	.019	.039	.070	1/2
3/32	.027	.054	.098	1/2
1/8	.040	.070	.124	1/2
5/32	.049	.086	.152	1/2
3/16	.057	.101	.180	1/2
1/4	.077	.133	.233	5/8

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3.8.3.1 Style A, short shank. The type II, class 3, style A rivet set shall have a cupping for 1/16, 3/32, and 1/8 inch size rivets as specified (see 6.2), and shall conform to the applicable dimensions shown on figure 7 and in table V.

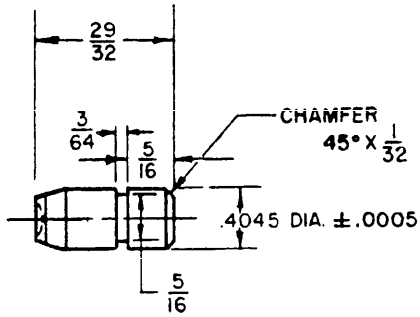


FIGURE 7. Type II, class 3, style A, short shank, universal-head rivet set.

3.8.3.2 Style B, Parker taper shank. The type II, class 3, style B rivet set shall have a Parker taper shank and a straight body, shape S, or an offset body, shape O, as specified (see 6.2), conforming to the dimensions shown on the applicable figure 8 or figure 9. The cupping shall be for the size rivet specified (see 6.2), and shall conform to the applicable dimensions shown in table V.

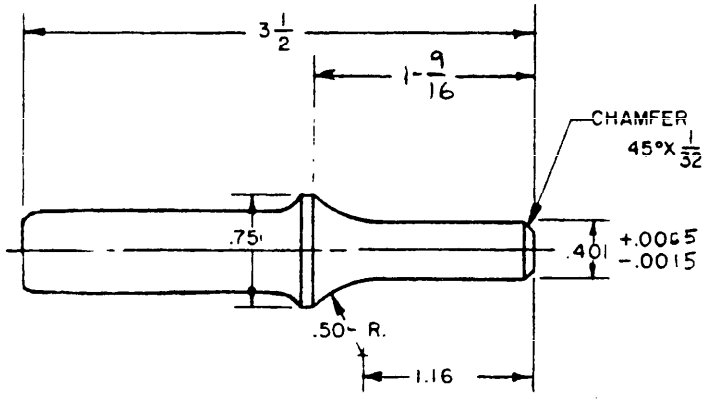


FIGURE 8. Type II, class 3 and class 6, style B, Parker taper shank, shape S, straight body rivet set.

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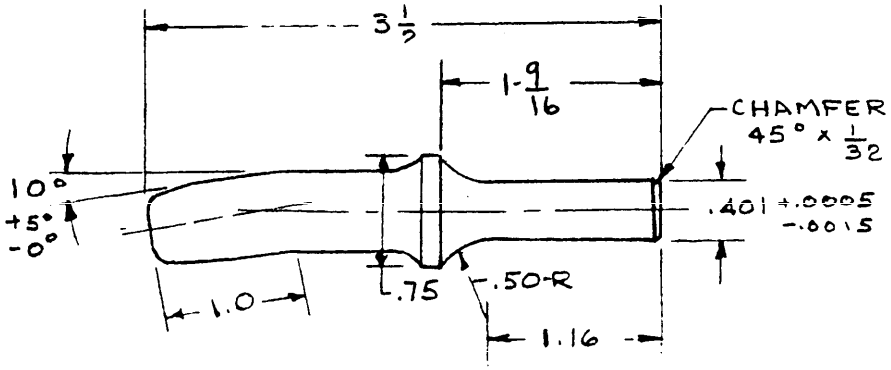


FIGURE 9. Type II, class 3 and class 6, style B, Parker taper shank, shape O, offset body rivet set.

3.8.4 Class 4, brazier head and class 5, modified brazier head. The type II, class 4 and class 5 rivet sets shall have a Parker taper shank and a straight body, shape S or an offset body, shape O, as specified (see 6.2), conforming to the dimensions shown on the applicable figure 8 or figure 9. The cupping shall be for the size rivet specified (see 6.2) and shall conform to the applicable dimensions on figure 10 and in table VI.

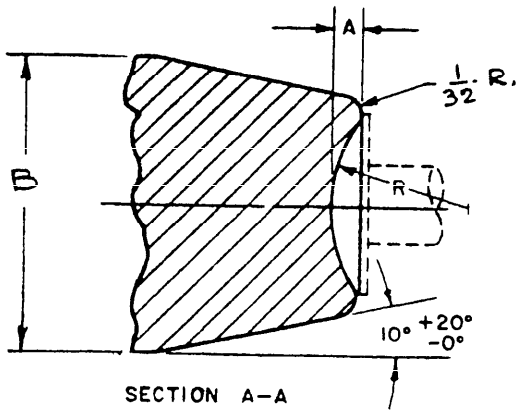


FIGURE 10. Type II, classes 4 and 5, brazier and modified brazier-head, rivet set.

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TABLE VI. Dimensions for type II, classes 4 and 5, brazier and modified brazier-head rivet sets

Class 4 rivet size	Class 5 rivet size	A Depth of cupping ±.002	R Radius of cupping ±.002	B Body diameter
	3/32	.018	.142	1/2
3/32	1/8	.030	.213	1/2
1/8	5/32	.047	.284	1/2
5/32	3/16	.057	.354	1/2
3/16	1/4	.069	.415	5/8
1/4		.094	.567	3/4

3.8.5 Class 6, round head. The type II, class 6, style B rivet set shall have a Parker taper shank and a straight body, shape S, or an offset body, shape O, as specified (see 6.2), conforming to the dimensions shown on the applicable figure 8 or figure 9. The cupping shall be for the size rivet specified (see 6.2), and shall conform to the applicable dimensions shown on figure 11 and in table VII.

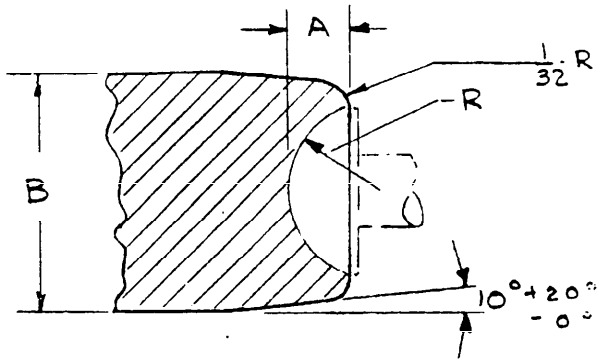


FIGURE 11. Type II, class 6, round head rivet set.

TABLE VII. Dimensions for type II, class 6, round head rivet sets

Rivet size	A Depth of cupping ±.002	R Radius of cupping ±.002	B Body diameter
1/16	0.031	0.075	1/2
3/32	.049	.112	1/2
1/8	.068	.161	1/2
5/32	.086	.184	1/2
3/16	.104	.222	1/2
1/4	.139	.298	5/8

3.9 Workmanship. The rivet set shall be free of any defect that may adversely effect the operation, serviceability, or life expectancy.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.2 Sampling for examinations and tests.

4.2.1 Inspection lot. An inspection lot shall consist of all rivet sets of one type, class, style, shape, and size offered for acceptance at one time. The sample unit shall be one rivet set.

4.2.2 Sampling for examinations and tests. Sampling for all examinations and tests shall be accomplished in accordance with MIL-STD-105 at inspection level II with an Acceptable Quality Level (AQL) of 4.0.

4.3 Examinations.

4.3.1 Each sample selected in accordance with 4.2.2 shall be examined for conformance to the requirements for dimensions, finish, markings, and workmanship. Any sample that does not comply with all requirements shall be classified as a reject.

4.3.2 Examination of preparation for delivery. Examination shall be made to determine compliance with preservation, packaging, packing, and marking requirements in accordance with PPP-P-40.

4.4 Tests.

4.4.1 Hardness test. Each sample shall be tested for the hardness specified in 3.7 and 3.8. Test impressions shall be taken at any location on the body, shank, and cuppings. Any sample that does not conform shall be classified as a reject.

4.4.2 Surface roughness test. Each sample shall be tested for surface roughness in accordance with ANSI B46.1 to determine compliance with the requirements of 3.4. Any sample that does not conform shall be classified as a reject.

4.4.3 Performance test. Each sample rivet set shall be tested under practical service conditions by setting at least 10 steel rivets of the size specified for the sample rivet set. The type I, class 1 rivet sets shall be tested by setting 3/8 inch size rivets. After setting rivets, the rivet set shall be examined and any evidence of distortion, abnormal wear, or other defect will classify that rivet set as a reject.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. The rivet sets shall be preserved, packaged, packed, and marked in accordance with PPP-P-40. Preservation, packaging, and packing shall be level A, B, or C, as specified (see 6.2), and the marking shall be civil or military agency as specified (see 6.2).

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6. NOTES

6.1 Intended use. Rivet sets covered by this specification are intended for use with chipping hammers and riveters covered by 00-H-116 and 00-R-421, respectively.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, style, and shape of rivet set required (seel.2).
(c) Size required (see 1.2.2).
(d) Applicable levels of preservation, packaging, and packing required and marking required (see 5.1).

6.3 Cross reference data. The offset body shape (Shape O) rivet sets were added to the class 3, style B and the class 6, style B rivet sets in this specification. The class 3, style B and the class 6, style B rivet sets in GGG-R-425a are now classified as Shape S, straight body shape.

MILITARY CUSTODIANS:

DOD has waived coordination on revisions and amendments to this Federal Specification until further notice.

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 15 cents each.

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